

Reliable Extraction from Epilepsy Letters

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Index Terms—component, formatting, style, styling, insert.

I. INTRODUCTION

A. Medical framing

Epilepsy is one of the most common neurological diseases 30 percent of patients are drug resistant and research progress has been stubbornly slow

We need a deeper understanding of who these patients are, how they are treated, how outcomes differ over time and across groups

The most important clinical markers are trapped in unstructured notes

Costly chart review is required for constructing patient timelines, identifying research cohorts, conducting retrospective studies

They struggle through that anyway because it's so valuable, but it's not just costly it's error prone Entity extraction vs. entity relational extraction and reasoning NLP solved entity extraction a while back, but this is a temporal and clinical reasoning problem - how to represent a clinical state in a rigorous way

It's not just computers that struggle with this problem, but doctors too - they all describe things differently and experts struggle to interpret them identically

B. Prior system summary

Rules-based systems are good at entity extraction but require extensive annotation and perform poorly on logic

BERT-based systems require extensive annotations and again, perform poorly on logic

GPT-based systems are very semantically flexible and strong on reasoning, but they're a bit too flexible - hallucination, rambling prose, it's unstructured data to unstructured data

Monolithic LLMs provide the least visibility into the most dangerous task, clinical reasoning

Most of the LLMs tested and performing well as monolithic systems were closed-source, which creates privacy concerns

These problems lead to systems generalising poorly to new tasks and styles We need the modularity and reliability of rules-based systems to constrain the flexibility of the LLMs and audit their outputs

C. Hybrid solution

I present a modular, evidence-grounded LLM framework for clinical information extraction from epilepsy clinical letters

It decomposes extraction into configurable stages for semantic interpretation, relation extraction, temporal resolution, concept normalisation, schema validation and evidence verification

It is evaluated across two independently developed, publicly shareable epilepsy benchmarks assessing two different capabilities: broad phenotype extraction in ExECTv2 and temporally complex seizure-frequency extraction from synthetic leets

By directly comparing rules-only, LLM-only and hybrid configurations, plus component ablations and error analysis, we show how modular, evidence-grounded hybrid systems not only improve field-level F1 scores but also reliability, auditability and generalisability

II. FIGURES AND TABLES

a) Positioning Figures and Tables: Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation "Fig. 1", even at the beginning of a sentence.

Table Head	Claim it supports	Evidence shown	Reviewer risk
Table 1: Dataset Summary	Method is robust and generalisable	Dataset size, labels, splits, class balance	How valuable is synthetic data?
Table 2: Main Results	Method improves over baseline in 3 key ways	F1, evidence, schema validity, uncertainty	What do schema and evidence validity mean?
Figure 1: Architecture	Method is modular, transparent, reproducible, and clinically logical	Pipeline stages, representative example	Architecture may seem unnecessarily convoluted
Table 3: Ablation	Deterministic and LLM components are complementary	LLM-only, rules-only, LLM + schema validation, LLM + evidence verifier, LLM + concept normalisation, LLM + temporal resolver, full hybrid	Unclear separation between rules and LLM; unclear rules
Figure 2: Error Analysis	Deterministic and LLM components have distinct error profiles that you can calibrate for	Hallucinations, low recall, schema validity, evidence coverage, normalisation errors, temporal contradiction, abstention rate	Difficulty identifying the most meaningful errors

Figure

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”,

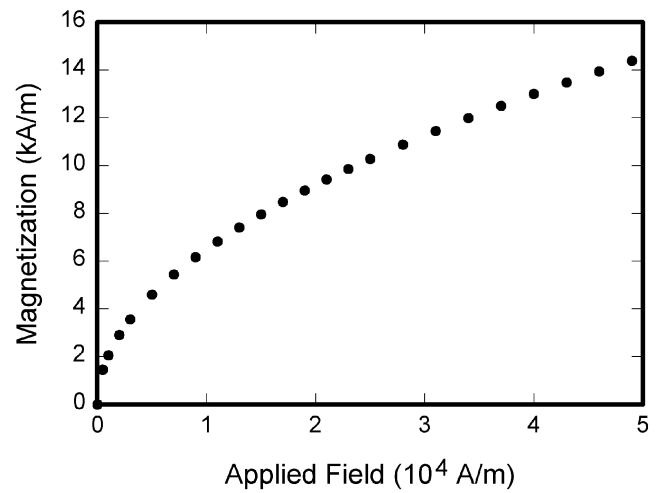


Fig. 1. Example of a figure caption.

not “Temperature/K”.

ACKNOWLEDGMENT

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REFERENCES

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REFERENCES

- [1] G. Eason, B. Noble, and I. N. Sneddon, “On certain integrals of Lipschitz-Hankel type involving products of Bessel functions,” *Phil. Trans. Roy. Soc. London*, vol. A247, pp. 529–551, April 1955.
- [2] J. Clerk Maxwell, *A Treatise on Electricity and Magnetism*, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.
- [3] I. S. Jacobs and C. P. Bean, “Fine particles, thin films and exchange anisotropy,” in *Magnetism*, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.

- [4] K. Elissa, "Title of paper if known," unpublished.
- [5] R. Nicole, "Title of paper with only first word capitalized," J. Name Stand. Abbrev., in press.
- [6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, "Electron spectroscopy studies on magneto-optical media and plastic substrate interface," IEEE Transl. J. Magn. Japan, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetism Japan, p. 301, 1982].
- [7] M. Young, The Technical Writer's Handbook. Mill Valley, CA: University Science, 1989.
- [8] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, arXiv:1312.6114. [Online]. Available: <https://arxiv.org/abs/1312.6114>
- [9] S. Liu, "Wi-Fi Energy Detection Testbed (12MTC)," 2023, gitHub repository. [Online]. Available: <https://github.com/liustone99/Wi-Fi-Energy-Detection-Testbed-12MTC>
- [10] "Treatment episode data set: discharges (TEDS-D): concatenated, 2006 to 2009." U.S. Department of Health and Human Services, Substance Abuse and Mental Health Services Administration, Office of Applied Studies, August, 2013, DOI:10.3886/ICPSR30122.v2
- [11] K. Eves and J. Valasek, "Adaptive control for singularly perturbed systems examples," Code Ocean, Aug. 2023. [Online]. Available: <https://codeocean.com/capsule/4989235/tree>

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